

Computer Vision – Lecture 1

Introduction

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Bastian Leibe

Visual Computing Institute

RWTH Aachen University

<http://www.vision.rwth-aachen.de/>

leibe@vision.rwth-aachen.de

Topics of Today's Lecture

- Motivation

- Why Computer Vision?
- What is Computer Vision?
- Challenges
- Overview over the Lecture

- Image Formation

- Historical development
- Pinhole Camera model
- Thin Lens model
- Camera parameters and their effect
- Digital images

Why Computer Vision?



Cameras are
all around us...



Images and video are everywhere...



Personal photo albums



Movies, news, sports



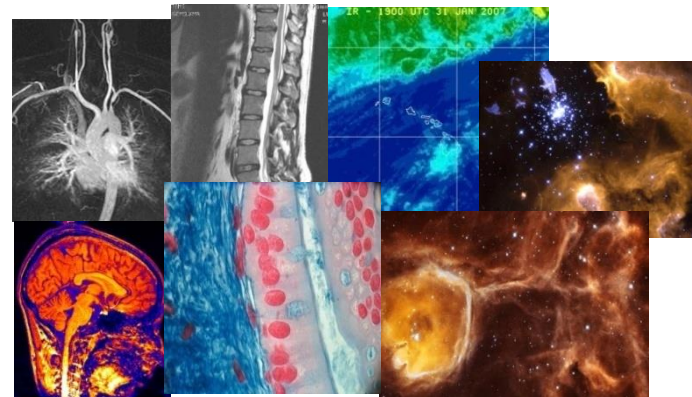
Internet services



Surveillance and security



Mobile and consumer applications



Medical and scientific images

What is Computer Vision?

- Goal of Computer Vision
 - Enable a machine to “*understand*” images and videos
- Automatic understanding
 - Computing properties of the 3D world from visual data (*measurement*)
 - Algorithms and representations to allow a machine to recognize objects, people, scenes, and activities. (*perception and interpretation*)

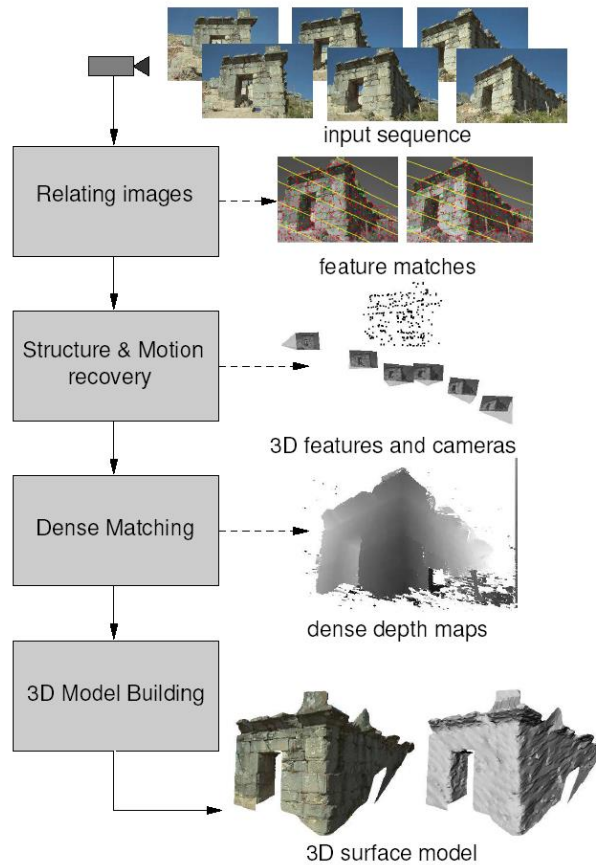
Vision for Measurement

Real-time stereo

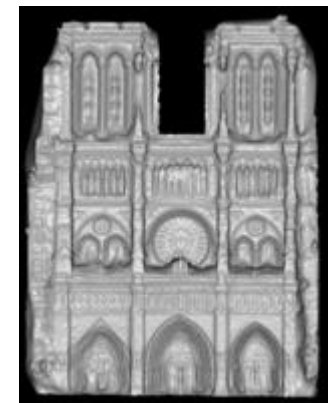


Pollefeys et al.

Structure from motion

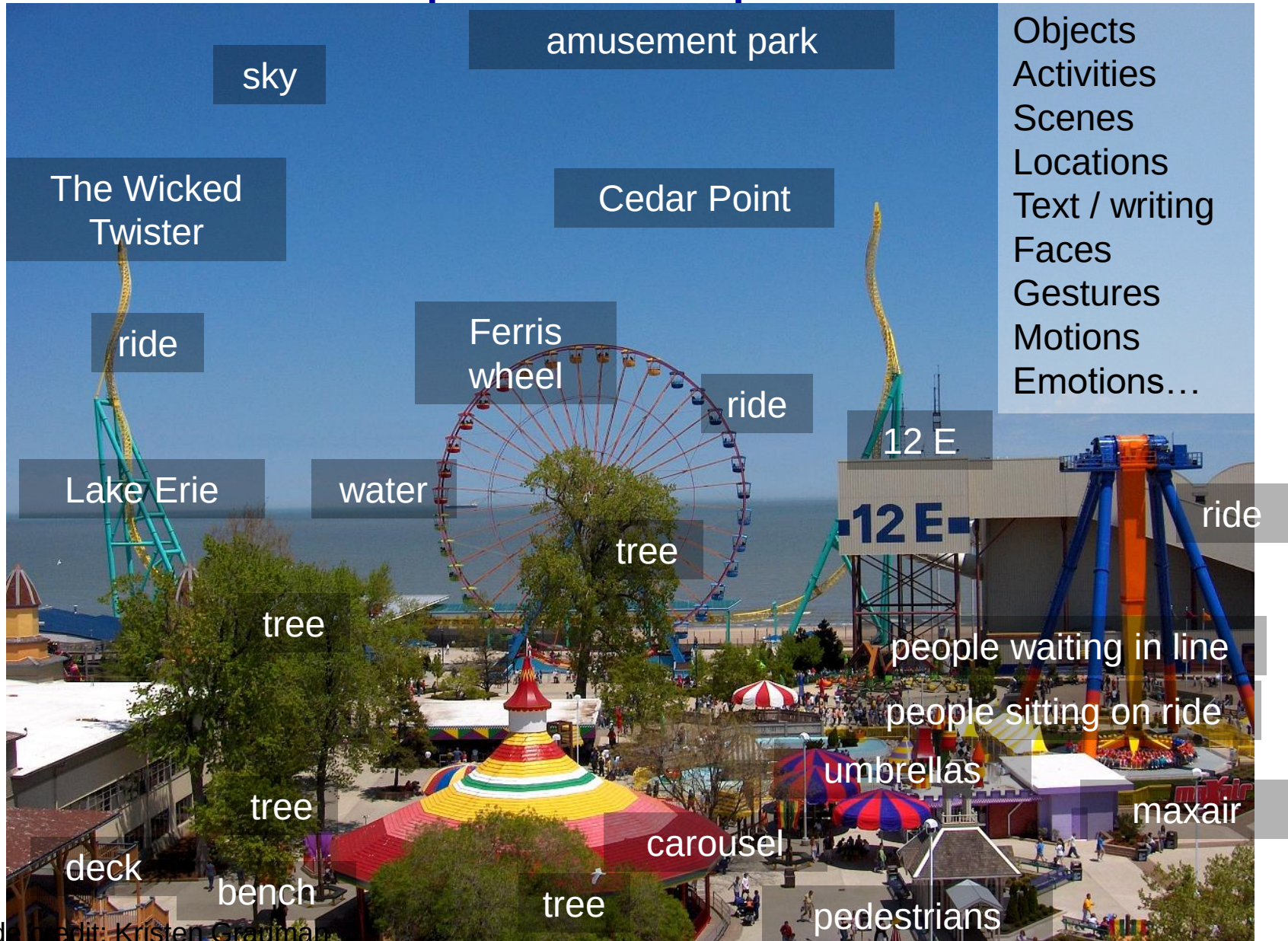


Multi-view stereo for community photo collections

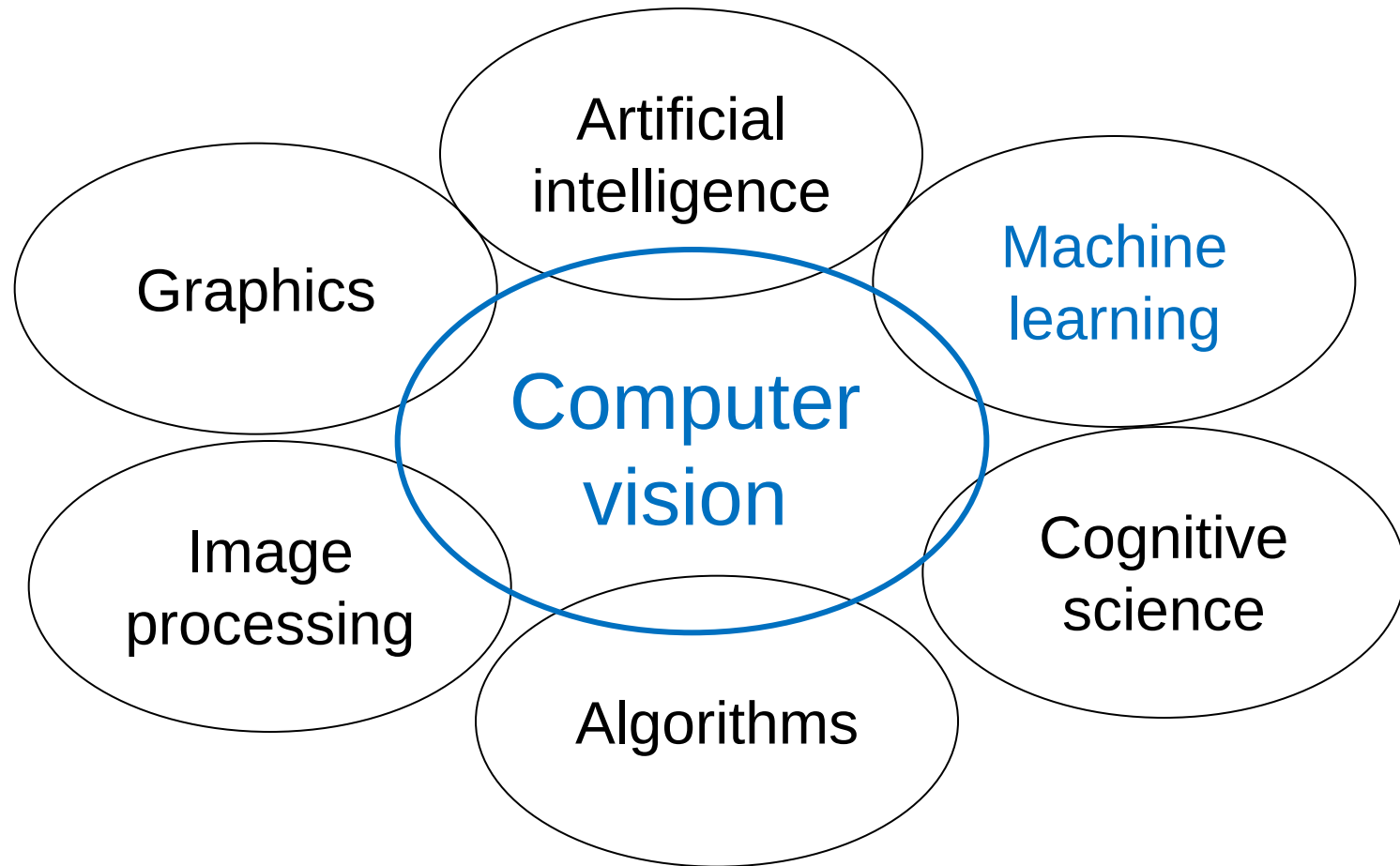


Goesele et al.

Vision for Perception, Interpretation



Related Disciplines



Directions to Computer Vision

- Science
 - Foundations of perception. How do WE see?
- Engineering
 - How do we build systems that perceive the world?
- Many applications
 - Medical imaging, automotive, entertainment, graphics, ...

Applications: Faces and Digital Cameras



Setting camera focus
via face detection



Camera waits for everyone to
smile to take a photo [Canon]

Automatic lighting
correction based on
face detection



Segmentation



- Automatic background removal from images
 - Functionality is included in Microsoft Office since 2010...



Matching



- Stitch your photos together to create panoramas

AUTOSTITCH



Applications: Vision for Mobile Phones

Google Goggles in Action

Click the icons below to see the different ways Google Goggles can be used.



- Take photos of objects as queries for visual search

Applications: Vision-based Interfaces



Games
(Microsoft Kinect)

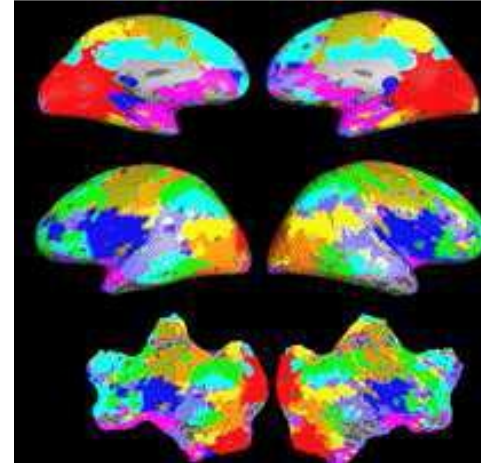


Assistive technology systems
Camera Mouse
Boston College

Applications: Medical & Neuroimaging



Image guided surgery
MIT AI Vision Group



fMRI data
Golland et al.



Applications: Visual Special Effects



The Matrix



MoCap for *Pirates of the Caribbean*, Industrial Light and Magic

(Source: S. Seitz)

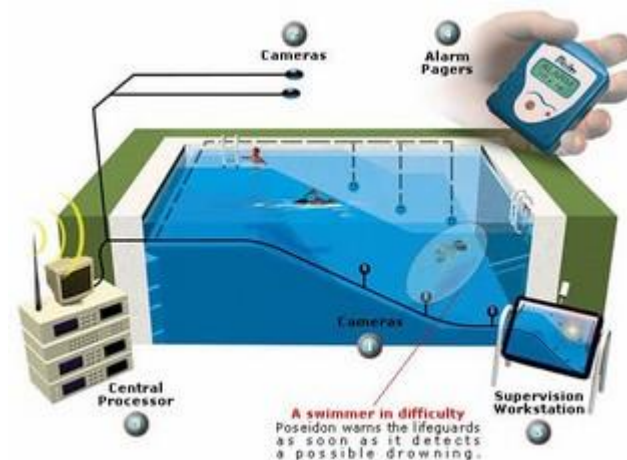
Applications: Safety & Security



Autonomous robots



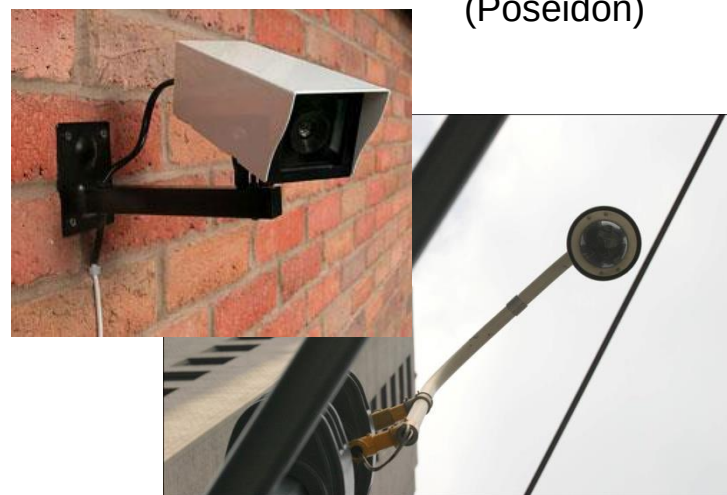
Driver assistance



Monitoring pools
(Poseidon)

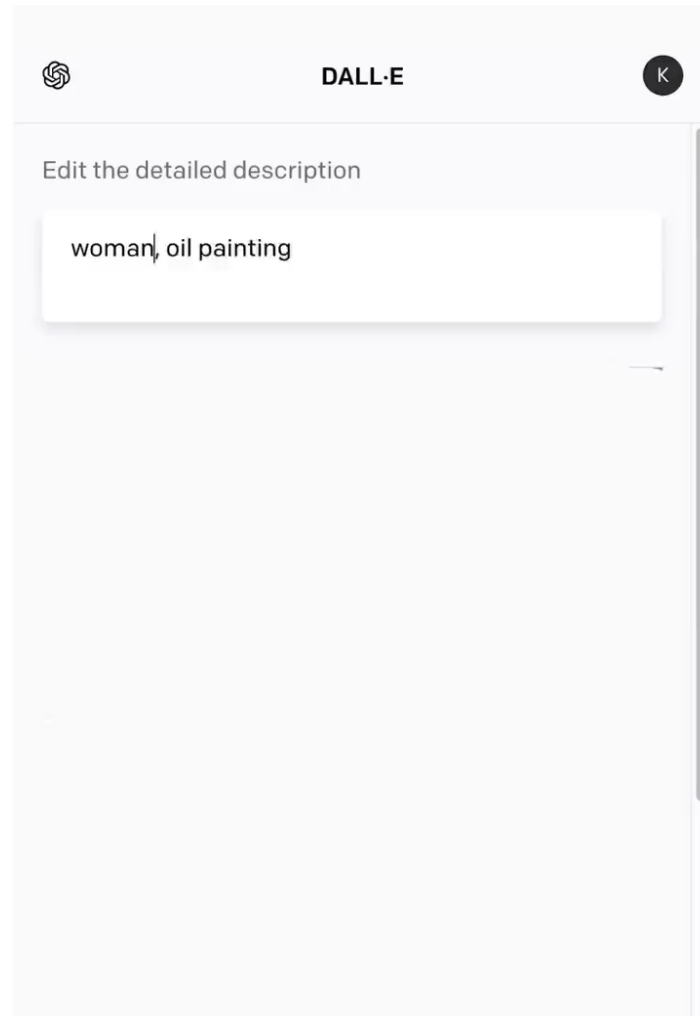


Pedestrian detection
[MERL, Viola et al.]



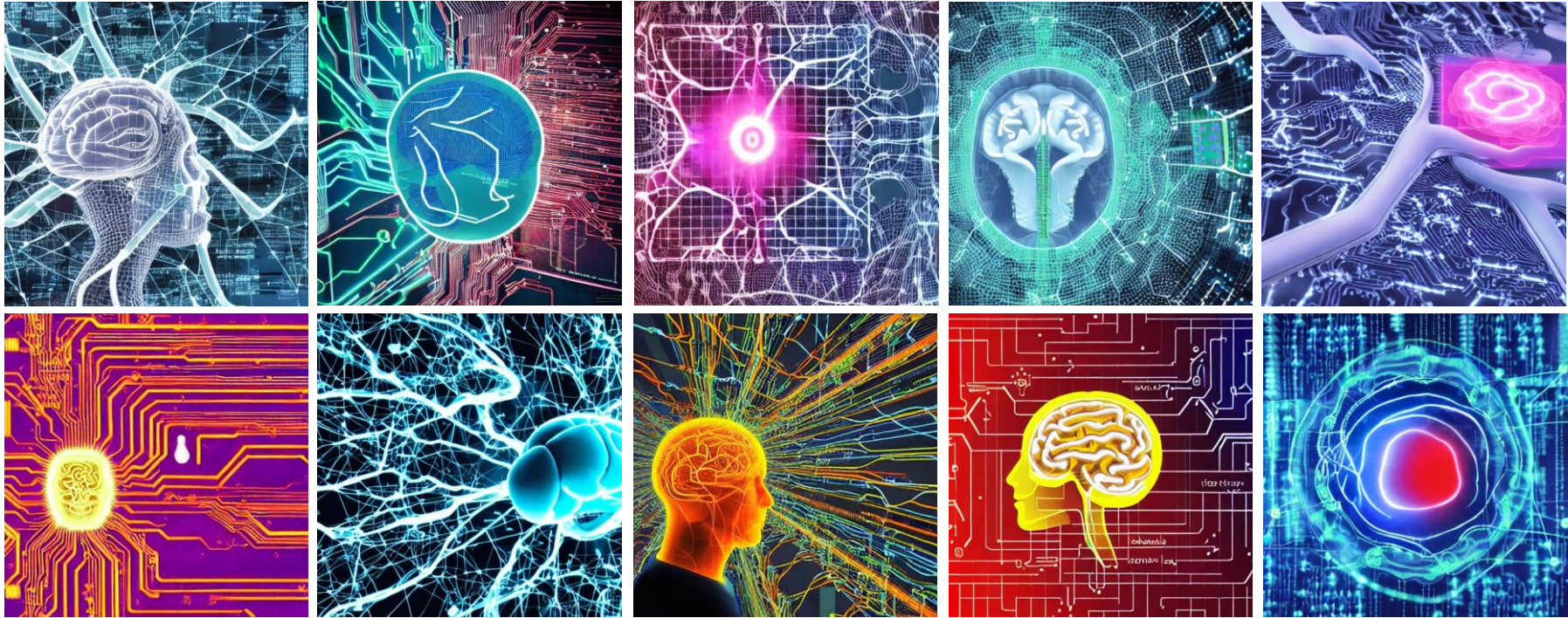
Surveillance

Applications: Image Generation Models



E.g., OpenAI's DALL-e

Very Exciting Times Are Ahead!

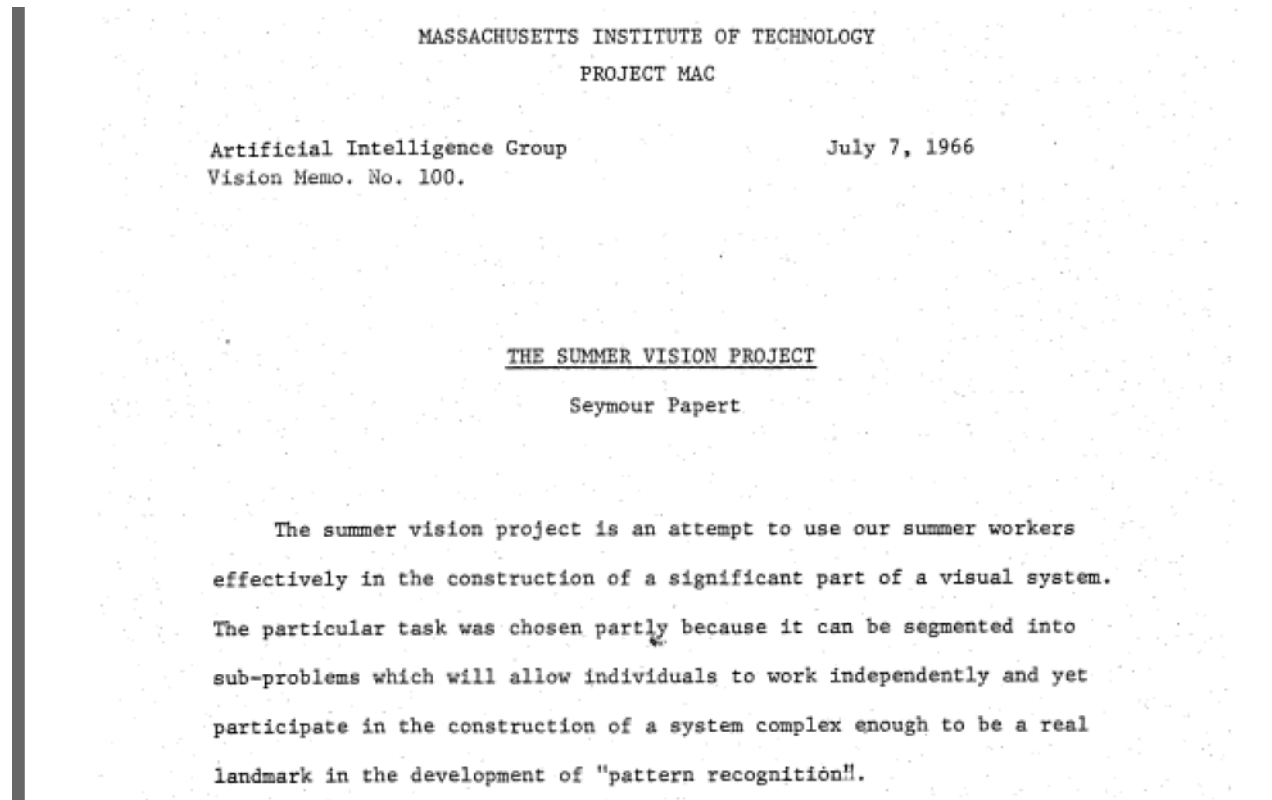


Images created with StableDiffusion by Stability.AI

- We're witness to the first strong AIs being created in front of our eyes...
 - ...and visual perception will be one of their core capabilities

Ok, Let's Do It – Any Obstacles?

- 1966: Seymour Papert directs an undergraduate student to solve "the problem of computer vision" as a summer project.



- Obviously, computer vision was too difficult for that...

Challenges: Many Nuisance Parameters



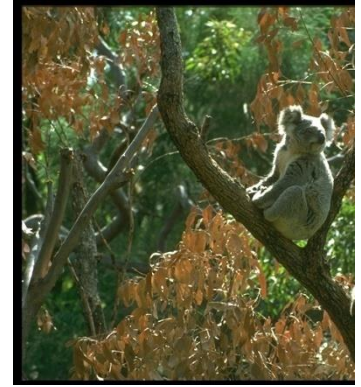
Illumination



Object pose



Clutter



Occlusions

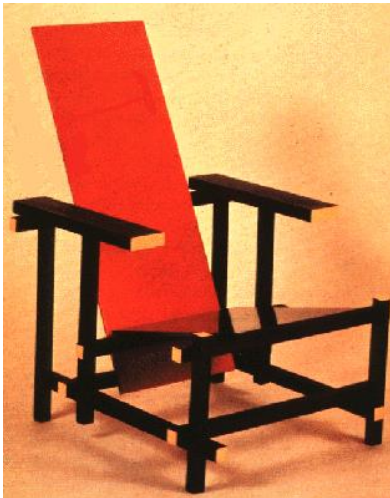


Intra-class
appearance



Viewpoint

Challenges: Intra-Category Variation

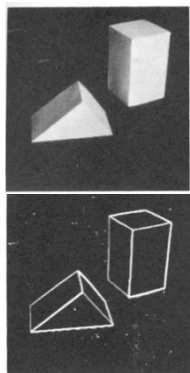


Challenges: Complexity

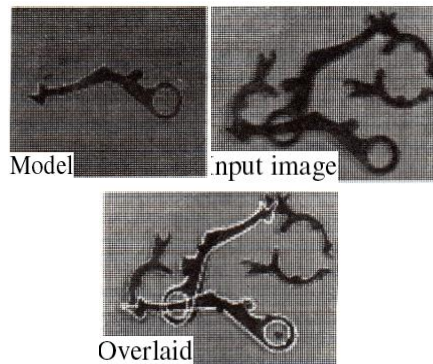
- Thousands to millions of pixels in an image
- 3,000-30,000 human recognizable object categories
- 30+ degrees of freedom in the pose of articulated objects (humans)
- Trillions of images indexed by Google Image Search
- Millions of hours of video available on youtube
- >1.5 billion smartphones sold worldwide in 2019, each equipped with at least 1 digital camera
- About half of the cerebral cortex in primates is devoted to processing visual information [Felleman and van Essen 1991].

So, Should We Give Up?

- NO! Very active research area with exciting progress!



...



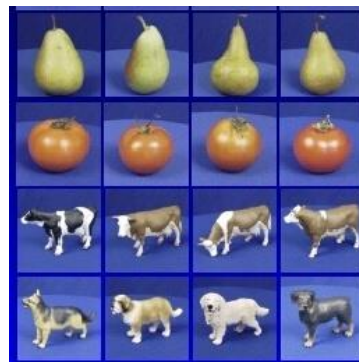
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Things Are Starting to Work...



Computer Vision in realistic scenarios is now feasible!

Things Are Starting to Work...



Computer Vision in realistic scenarios is now feasible!

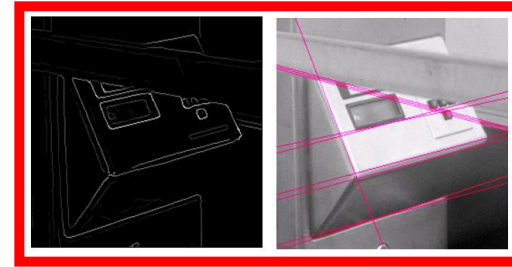
Things Are Starting to Work...



Computer Vision in realistic scenarios is becoming feasible!

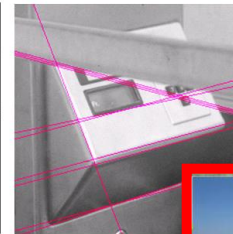
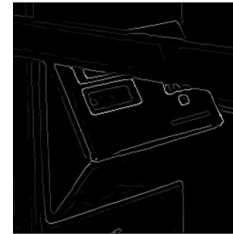
Course Outline

- Image Processing Basics
- Segmentation
- Object Recognition & Categorization
- Deep Learning Approaches (CNNs, Transformers)
- 3D Reconstruction



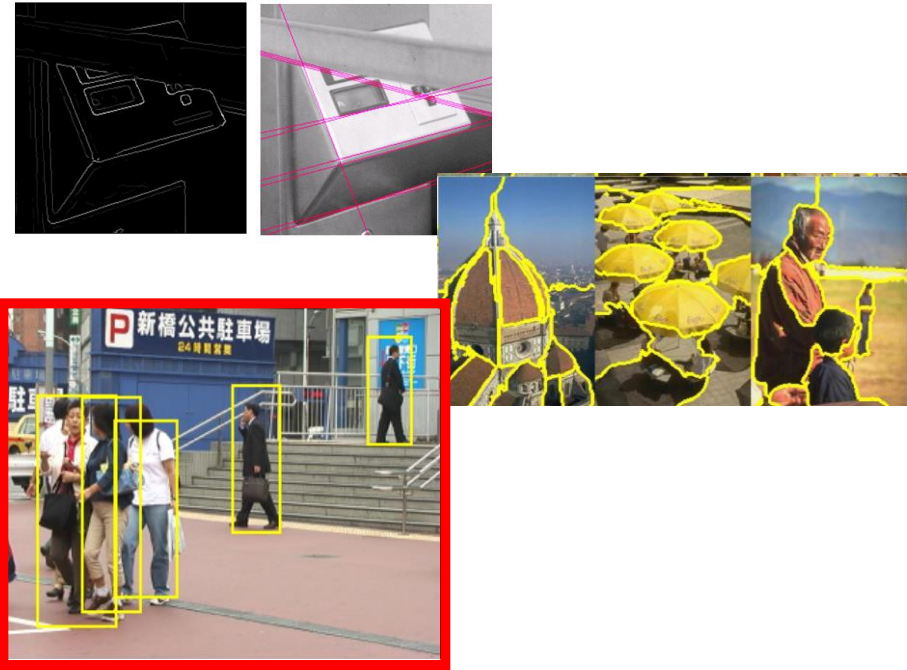
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And you might have heard of...



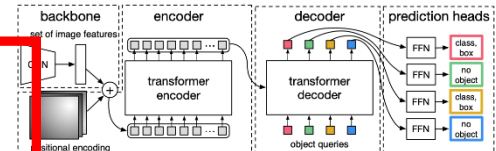
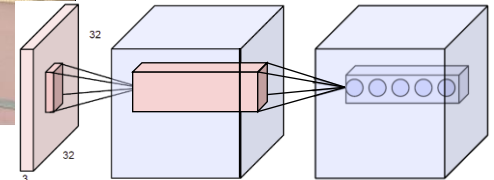
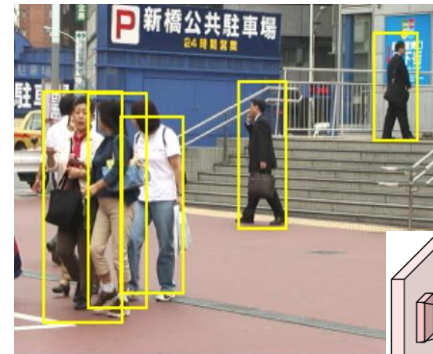
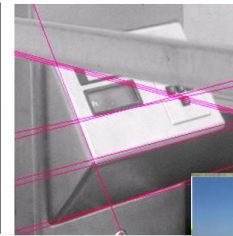
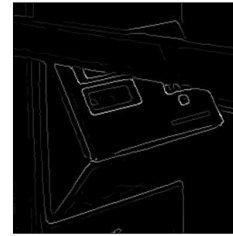
Deep Learning

- Image Processing Basics
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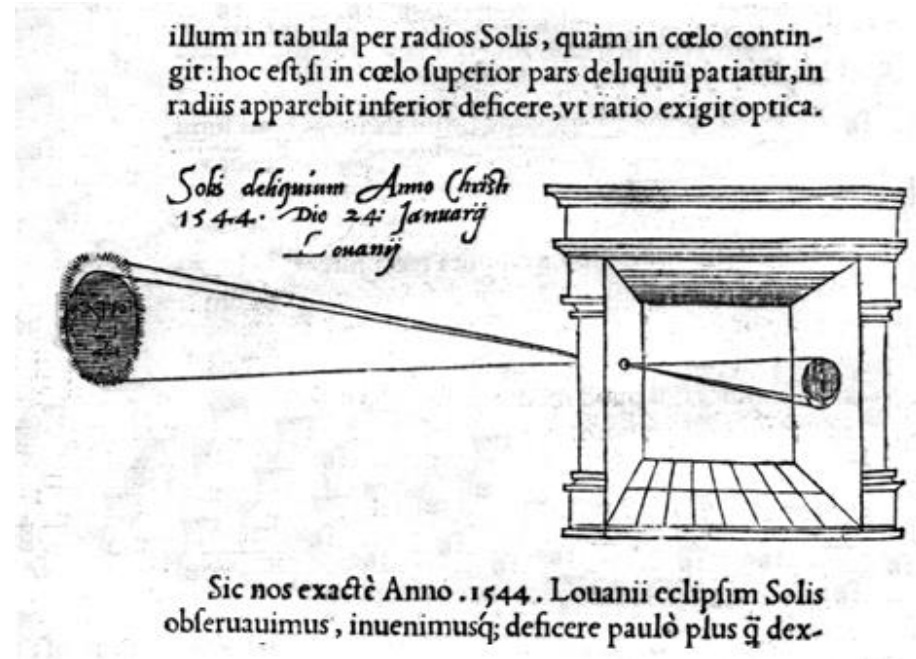
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 - Historical development
 - Pinhole Camera model
 - Thin Lens model
 - Camera parameters and their effect
 - Digital images

Camera Obscura

“When images of illuminated objects ... penetrate through a small hole into a very dark room ... you will see [on the opposite wall] these objects in their proper form and color, reduced in size ... in a reversed position owing to the intersection of the rays”

Leonardo Da Vinci



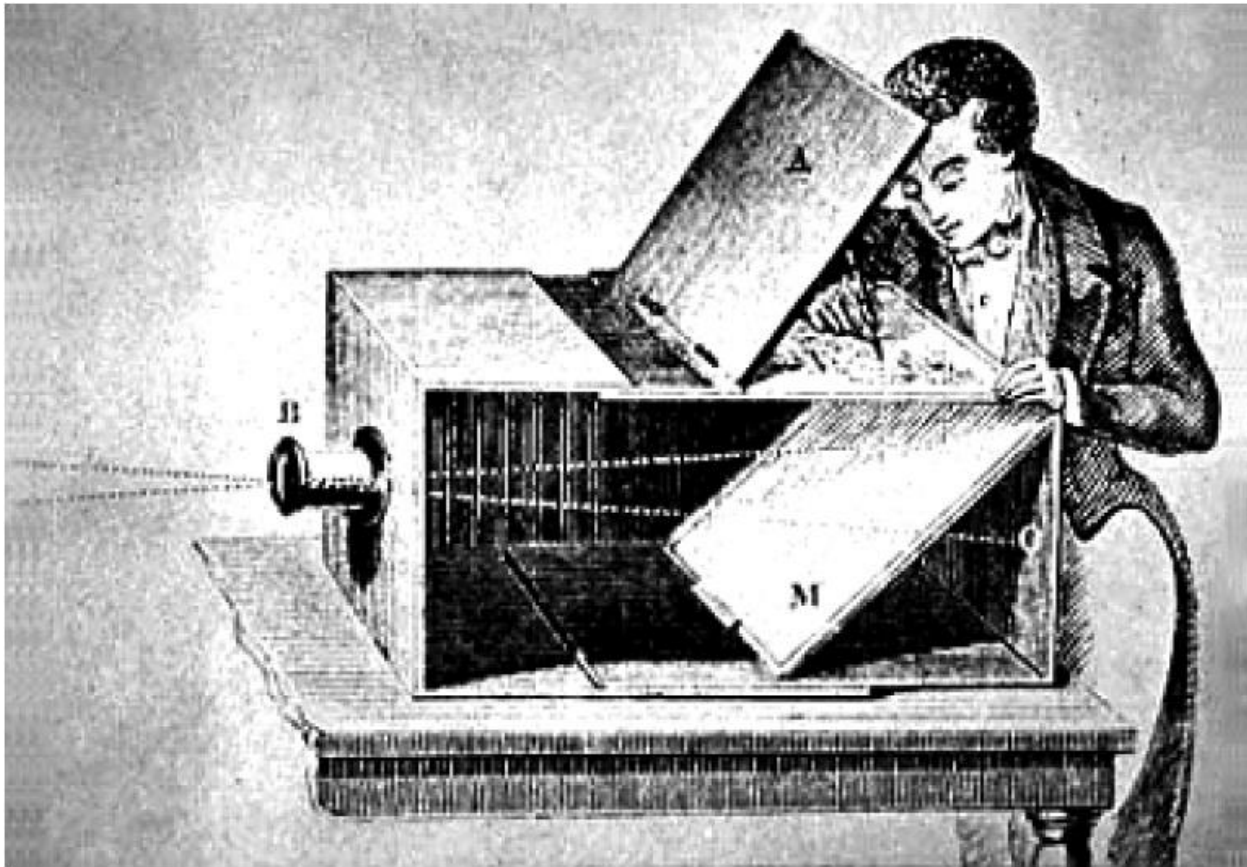
- Literally, “Dark Chamber”
 - Principle known for a long time
 - First written account by Alhazen (965 – 1038)
 - Intensely studied by Leonardo Da Vinci (1452 – 1519)
 - Used as a tool for studying solar eclipses

You Can Observe This Yourself!

- Gaps through tree leaves act as pinholes during a solar eclipse

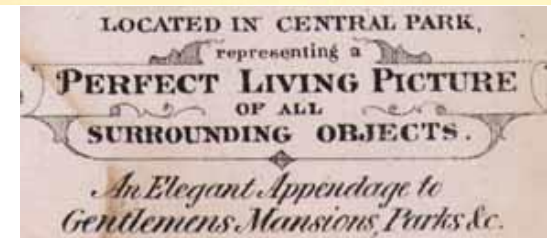


Camera Obscura



- Later on used by artists (e.g. Vermeer 17th century)

Camera Obscura



Jetty at Margate England, 1898.

An attraction in the late 19th century

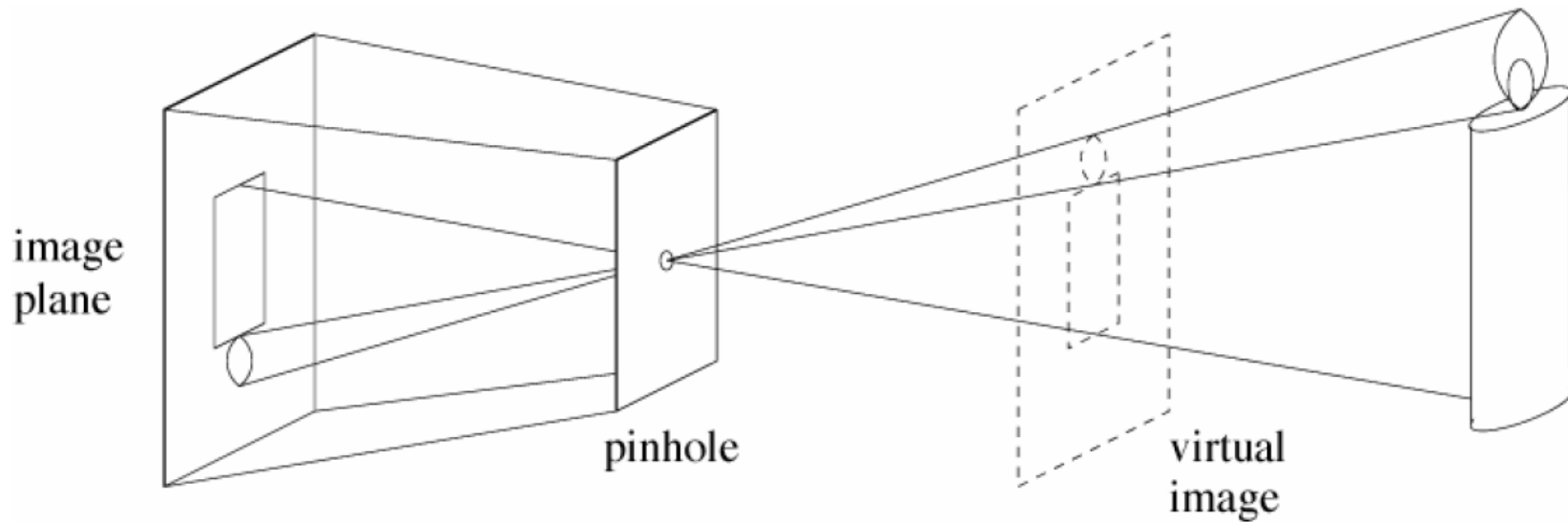
Camera Obscura



- Principle also used for lookout towers to monitor ship traffic
 - E.g., Torres miradores in Cadiz, Spain

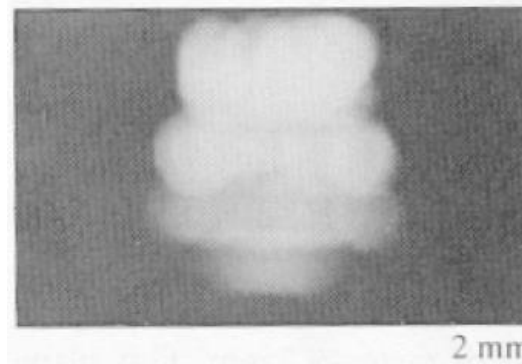
Pinhole Camera

- (Simple) standard and abstract model today
 - Box with a small hole in it
 - Works in practice



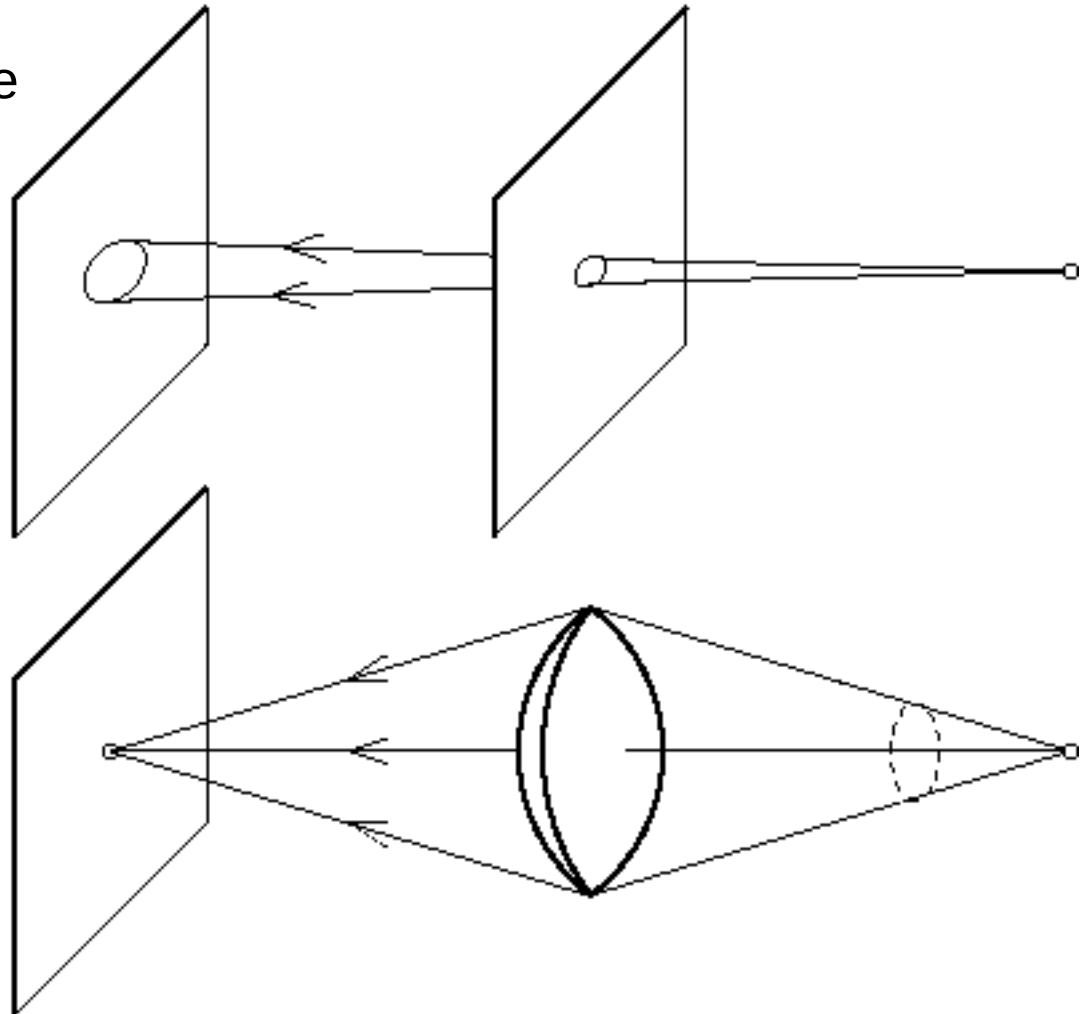
Pinhole Size / Aperture

- Pinhole too big – many directions are averaged, blurring the image
- Pinhole too small – diffraction effects blur the image
- Generally, pinhole cameras are *dark*, because a very small set of rays from a particular point hits the screen.

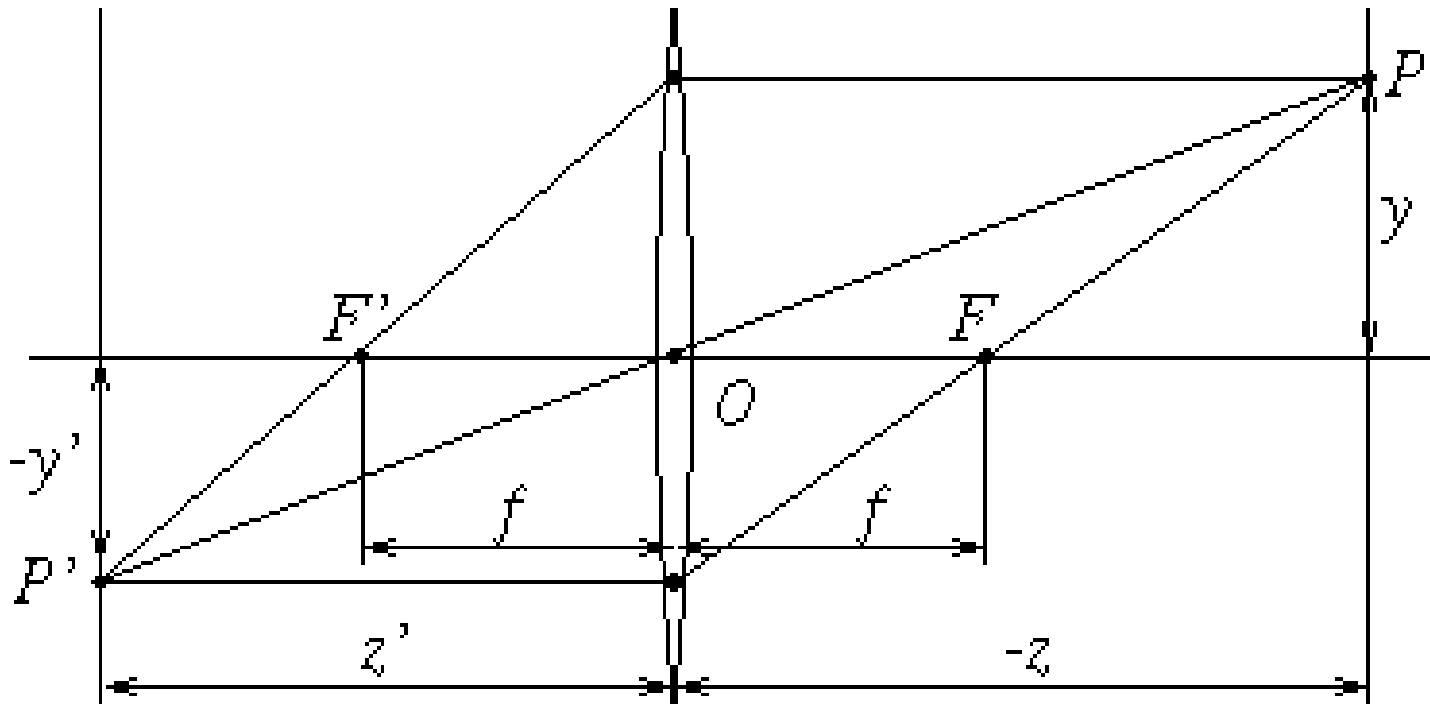


The Reason for Lenses

- Increasing the pinhole size to increase the amount of light causes blur.
- Lenses keep the image in sharp focus while gathering light from a larger area.



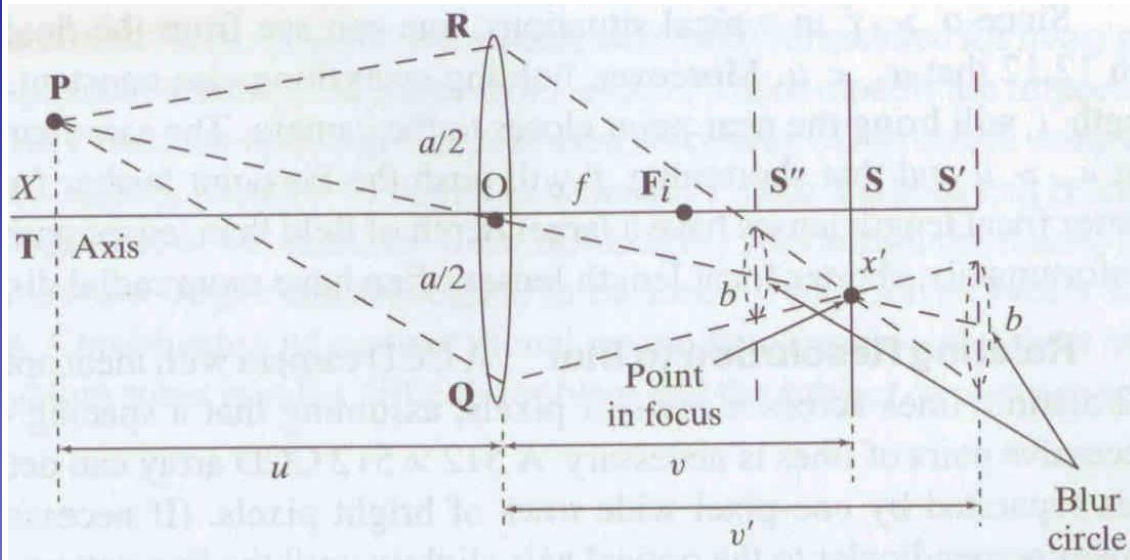
The Thin Lens



- Valid if the lens thickness is small compared to the radius of curvature

$$\frac{1}{z'} - \frac{1}{z} = \frac{1}{f}$$

Focus and Depth of Field



“circles of confusion”

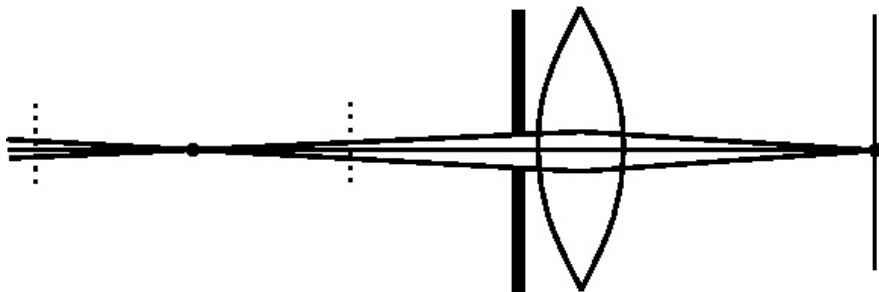
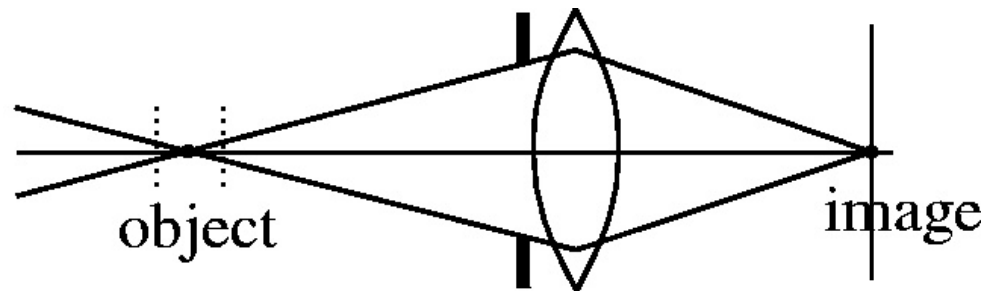
Thin lens: scene points at distinct depths come in focus at different image planes.

(Real camera lens systems have greater depth of field.)

- Depth of field: distance between image planes where blur is tolerable

Focus and Depth of Field

- How does the aperture affect the depth of field?

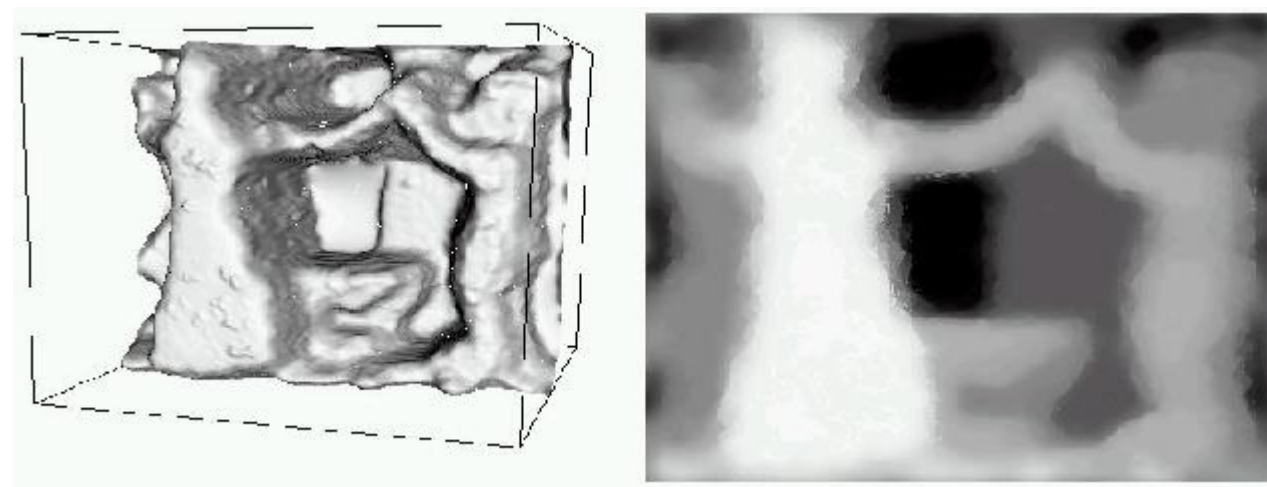


- A smaller aperture increases the range in which the object is approximately in focus

Application: Depth from (De-)Focus



Images from
same point of
view, different
camera
parameters



3D Shape /
depth
estimates

Field of View



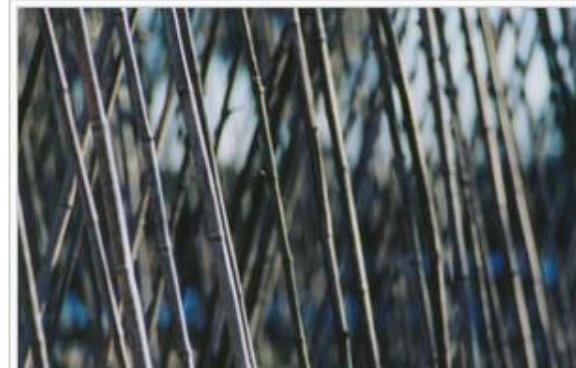
28 mm lens, $65.5^\circ \times 46.4^\circ$



50 mm lens, $39.6^\circ \times 27.0^\circ$



70 mm lens, $28.9^\circ \times 19.5^\circ$

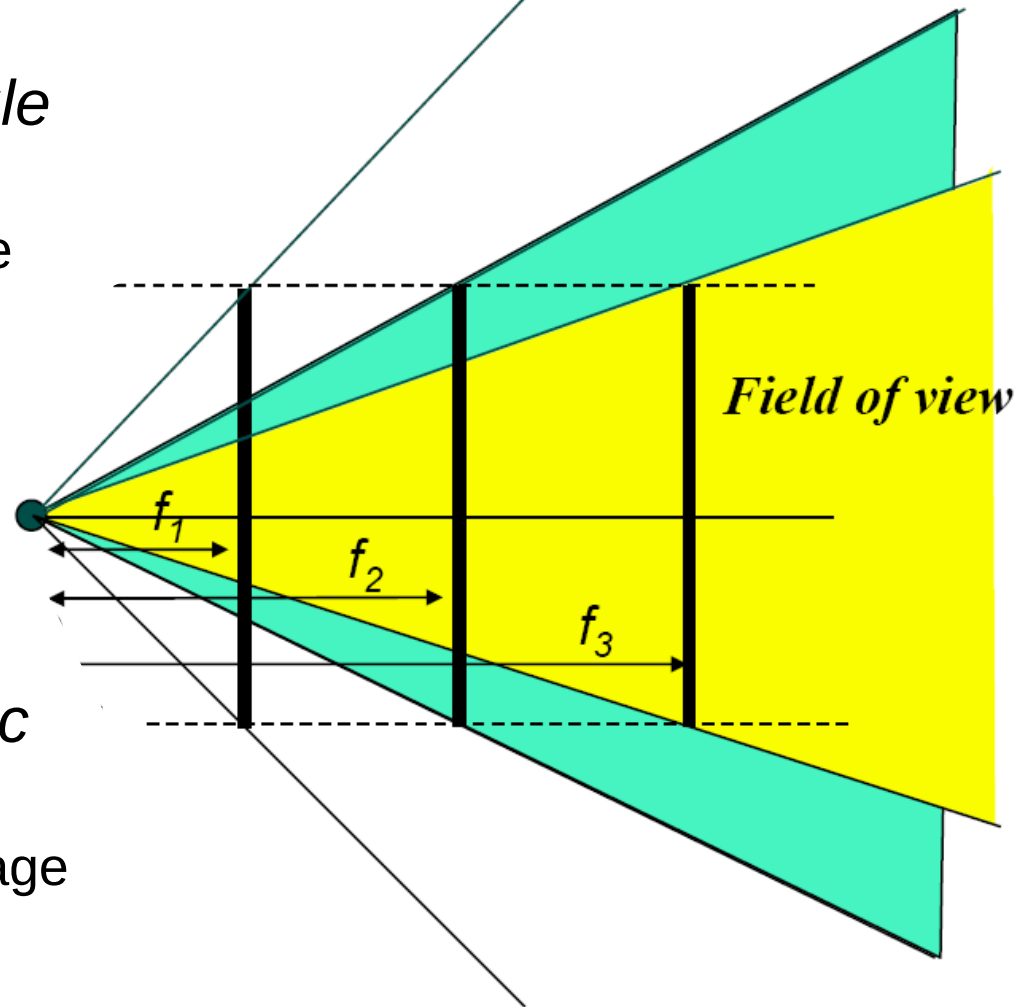


210 mm lens, $9.8^\circ \times 6.5^\circ$

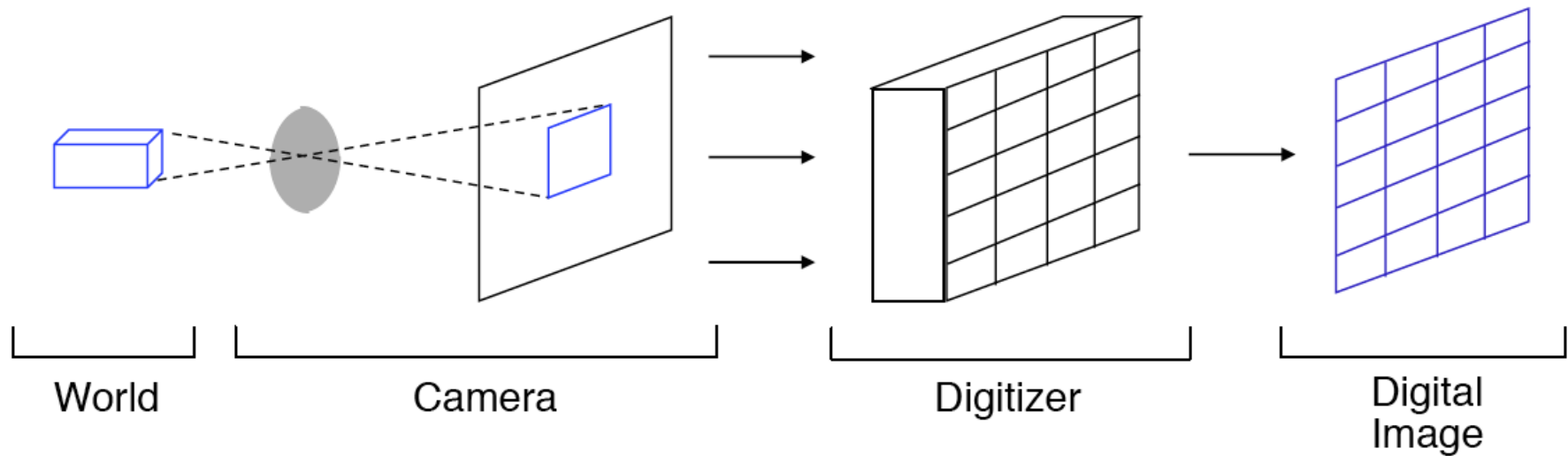
- Angular measure of the portion of 3D space seen by the camera

Field of View Depends on Focal Length

- As f gets smaller, image becomes more *wide angle*
 - More world points project onto the finite image plane
- As f gets larger, image becomes more *telescopic*
 - Smaller part of the world projects onto the finite image plane



Digital Images



- Film is replaced by a sensor array
- Current technology: arrays of *charge coupled devices* (CCD) or CMOS image sensors
- **Discretize** the image into pixels.
- **Quantize** light intensities into pixel values.

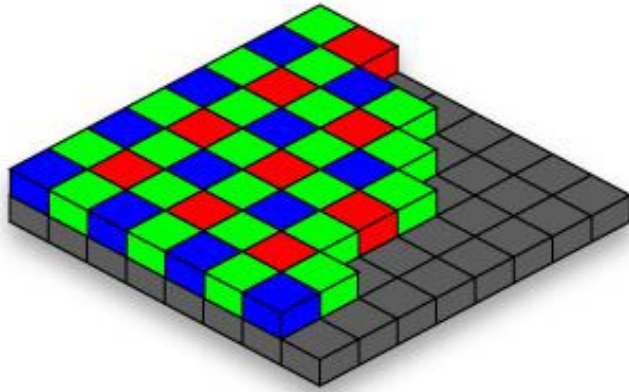
Resolution

- Sensor: size of real world scene element that images to a single pixel
- Image: number of pixels
- Influences what analysis is feasible, affects best representation choice

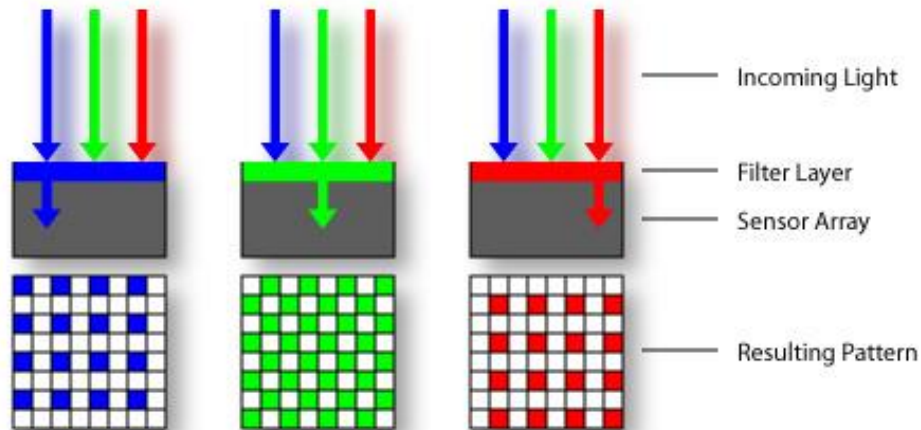


Color Sensing in Digital Cameras

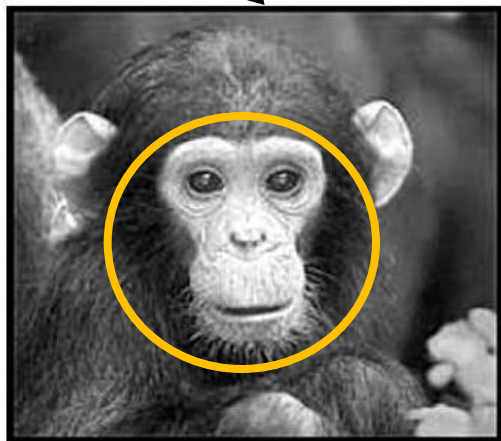
Bayer grid



Estimate missing components from neighboring values (**demosaicing**)



Color Images, RGB Color Space

**R****G****B**

Slide credit: Kristen Grauman

Grayscale Image

- Problem of Computer Vision

- How can we recognize fruits from an array of (gray-scale) numbers?
- How can we perceive depth from an array of (gray-scale) numbers?
- ...



		x =																
		58	59	60	61	62	63	64	65	66	67	68	69	70	71	72		
y =	41	210	209	204	202	197	247	143	71	64	80	84	54	54	57	58		
	42	206	196	203	197	195	210	207	56	63	58	53	53	61	62	51		
	43	201	207	192	201	198	213	156	69	65	57	55	52	53	60	59		
	44	216	206	211	193	202	207	208	57	69	60	55	77	49	62	61		
	45	221	206	211	194	196	197	220	56	63	60	55	46	97	58	106		
	46	209	214	224	199	194	193	204	173	64	60	59	51	62	56	48		
	47	204	212	213	208	191	190	191	214	60	62	66	76	51	49	55		
	48	214	215	215	207	208	180	172	188	69	72	55	49	56	52	56		
	49	209	205	214	205	204	196	187	196	86	62	66	87	57	60	48		
	50	208	209	205	203	202	186	174	185	149	71	63	55	55	45	56		
	51	207	210	211	199	217	194	183	177	209	90	62	64	52	93	52		
	52	208	205	209	209	197	194	183	187	187	239	58	68	61	51	56		
	53	204	206	203	209	195	203	188	185	183	221	75	61	58	60	60		
	54	200	203	199	236	188	197	183	190	183	196	122	63	58	64	66		
	55	205	210	202	203	199	197	196	181	173	186	105	62	57	64	63		

- What does it mean to see? How can we make a computer do it?

Next Lectures

- First few lectures: low-level vision
 - Filtering operations
 - Edge and structure extraction
 - Segmentation and grouping
- Next lecture: Linear Filters

References and Further Reading

- Background information on image formation can be found in Chapter 2 of the Szeliski book.

R. Szeliski
Computer Vision – Algorithms and Applications
Springer, 2010
([draft version available online](#))

